GCP AI-Native Document Operations Platform

Production-minded reference architecture for AI, cloud, APIs and business operations

Project objective

Build a cloud-ready document processing service that demonstrates how AI moves from a prototype into an API-first, controlled workflow. The platform separates extraction, AI classification, validation, routing, persistence and FinOps telemetry so each component can be tested and replaced independently.

Verified local run

The deterministic fixture contains 242 synthetic business documents. The workflow auto-processed 164, routed 78 to human review and rejected 0. The illustrative local cost estimator totalled USD 0.5992. All 20 automated tests passed.

Architecture

Client -> Apigee-ready API contract -> Cloud Run/FastAPI service -> Document AI extraction -> deterministic validation -> Vertex AI classification -> routing decision -> BigQuery audit/event store -> analytics and FinOps telemetry.

Engineering controls

- Typed Pydantic contracts and OpenAPI-compatible API design.
- Health/readiness endpoints, container packaging and GitHub Actions CI.
- Confidence, completeness and high-value review gates before straight-through processing.
- Structured audit persistence with latency and per-request cost telemetry.
- Terraform assets for API enablement, service account, BigQuery dataset, Artifact Registry and Cloud Run.
- Security notes covering least privilege, secret handling, logging, retention and data classification.

Truth boundary

The repository is fully runnable offline. Google Document AI, Vertex AI, BigQuery, Cloud Run and Apigee are represented by production seams, infrastructure/deployment assets and deterministic local adapters. They have not been executed against a live GCP project, so the CV and README do not claim live cloud deployment or cloud performance.

Why it is relevant

The project deliberately combines AI integration with API design, microservice-friendly stateless processing, infrastructure-as-code, CI/CD, structured data pipelines, human controls and cost instrumentation. It therefore demonstrates the path from business requirement and architecture through to a production-shaped engineering deliverable rather than a notebook-only AI demo.